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# Effect of insulating liquid viscosity on output force of a high-power synchronous electrostatic motor

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ABSTRACT

The output force of high-power electrostatic motors is defined by an internal electric field, and a strong electric field can cause air discharge unless the motors are operated within a dielectric liquid. Previous studies evaluated how the viscosity of a dielectric liquid can impact the output performance of a motor. While the viscous drag resistance increases with viscosity, the degradation of the output force was found to be negligible. However, the reason for this behavior was not clarified in those studies. This work clarifies the reason for the output behavior through numerical simulations and experiments. In the simulation, motion trajectories and motor output were analyzed for different operation speeds in different liquid viscosities. The results revealed that the existence of peaks in the force characteristics of the motor is the key to understanding a motor's unique behavior within a viscous liquid. In a viscous liquid, the trajectories change such that the peak force is generated to compensate for the drag resistance. The simulations also revealed that the behavior only appears in one of the two types tested in the simulation, which differ in the smoothness of their force characteristics. The simulation results were verified in experiments. The experimental results reconfirmed the results of previous studies, and verified the findings of the simulations of this work.

## 1. Introduction

Recently, soft actuators have been gathering attention for robotic applications [1–4]. Soft actuators are expected to realize safe contact in various environments and to increase affinity with human users. Thus, these devices would be useful for assistance robots, medical robots, and bio-inspired robots. The various types of soft actuators [5,6] include pneumatic, hydraulic, shape memory alloy, coiled polymer, PVC gel, and electrostatic actuators. While soft actuators exhibit unique mechanical flexibility, they typically fail to satisfy some of the conventional requirements in robotics regarding output force, motion range, operation speed, response frequency, overall system size, and control performance. For example, McKibben actuators [7], which are pneumatically operated, have a flexible shape and can exert a large force. However, they need a large air compressor and an air tank, which makes the total system much larger than conventional actuators [8]. The hydraulic actuator also has the same problem that the entire system is large [9,10]. As other examples, coiled polymer actuators [11] show a large force density, but the response is limited due to an inherent long cooling time [12]. In addition, PVC gel can be driven to produce a large output force with the same type of power supply devices as that for an electromagnetic motor, but the displacement is inherently small [13,14].

This work focuses on electrostatic actuators, as they have the potential to satisfy most of the requirements for high-power soft actuators [15]. As electrostatic actuators are driven electrically, they do not need compressors or tanks. Their motion principles are analogous to those of electromagnetic motors, and therefore, similar control strategies can be used to realize good control performance. Although the majority of the reported electrostatic actuators have been realized at micrometer scales as MEMS devices [16–18], many macroscopic actuators can produce sufficient force and torque [19–27]. These actuators require a high voltage, which is a drawback, but miniaturized multiplier drivers have been developed by reviewing and revising voltage circuits [28,29] and switching elements [30,31], and open-source miniature drivers are now available [32]. Therefore, the possibility of a solution is now well in sight. Furthermore, some electrostatic actuators are not only strong but also mechanically flexible, like human muscles [19–22]. Therefore, some electrostatic actuators that combine

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**Nomenclature**

| | |
|---|---|
| $\eta$ | Dynamic viscosity |
| $\mu$ | Friction coefficient |
| $\phi$ | Voltage phase difference between stator and slider |
| $\phi_{st}/\phi_{sl}$ | Voltage phase angle for stator/slider |
| $\theta_x$ | Electric angle of slider displacement |
| $C_x$ | Capacitance amplitude, $x$ = m0, m1, m2, m3, t3, l3 |
| $d$ | Damping coefficient |
| $F_l$ | External load force |
| $F_x$ | Thrust force between films |
| $F_y$ | Attraction force between films |
| $F_{\mu 0}$ | Friction force between films |
| $f_{st}/f_{sl}$ | Voltage frequencies for stator/slider |
| $k$ | Spring coefficient |
| $m$ | Slider mass |
| $p$ | Electrode pitch |
| $S$ | Stator and slider overlap area |
| $u$ | Slider moving speed |
| $V$ | Voltage amplitude |
| $v_s$ | Synchronous speed |
| $x$ | Slider displacement |
| $y_0$ | Gap between films |

a high level of strength and flexibility [19,20] seem to be promising candidates for robotic applications, especially for tasks in human environments. In these tasks, conventional robots have attempted to achieve safe contact with people and the environment through torque control [33,34] and flexible exteriors [35,36], but have been unable to do so adequately. However, by using a simple system that takes advantage of the flexibility of the actuator itself, safe contact should be possible.

The output force (torque) of electrostatic actuators is mostly defined by the electric field strength within the gap between the stator and the slider/rotor. As seen in the definition of the Maxwell stress for an electric field, the force is proportional to the square of the electric field. Therefore, a large field must be created in the gap for high performance. This, however, causes a discharge problem, especially in the air; strong fields can cause sparks or corona discharges into the air [37], which must be avoided for stable operation.

According to Paschen's law [38], discharges are not likely to occur in a micrometer-scale gap, and this property is fully used in MEMS actuators [17,18]. However, in case of large electrostatic actuators, use of a micro-scale gap is not practical. Therefore, a way to prevent air discharges in the gap is required. Filling the gap with an elastomer, as in a dielectric elastomer actuator, is one solution [39]. However, this solution limits the motion stroke considerably. It is possible to increase the displacement by revising the structure [23,24] or the elastomer material [25,40], but this creates another problem in that the deformation becomes nonlinear and cannot be easily controlled. Other solutions are filling the gap with media having a high withstand voltage, such as a high vacuum [41] and dielectric liquids. A high-vacuum gap is not practical in most cases, because it requires additional components, such as vacuum pumps. Therefore, the use of dielectric liquid is preferred to develop high-power electrostatic actuators [42].

Examples of high-performance actuators using dielectric liquid include, for example, linear film motors [20,43], the HASEL actuator [19], stacked electrostatic actuators [27,44], and large-torque rotary motors [26]. Linear film motors recorded an extremely large output force, such as 310 N, by operating within a dielectric liquid [20].

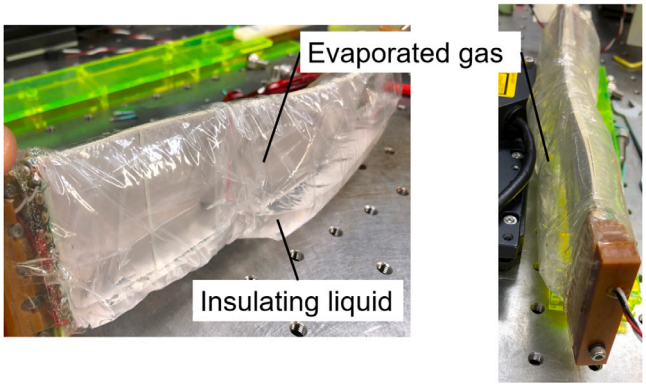



**Fig. 1.** Encapsulated electrostatic linear motor containing insulating liquid. The liquid left at room temperature (roughly around 25 °C) evaporates to inflate the bag like a balloon.

The HASEL actuator using an insulating liquid realized a high energy density by zipping actuation as a hydraulic drive source [19,45]. Stacked electrostatic actuators are driven in an insulating liquid with a spring-like structure of parallel plate electrodes to achieve a large displacement [27,44]. Ge et al. evaluated several types of dielectric liquids for a rotary electrostatic motor [26], in which they realized a large torque by using a liquid with moderate permittivity. However, it was shown that during operation, the driving force is greatly reduced due to the viscosity of the insulating liquid [46].

In this study, among the electrostatic actuators described above, we focused on the linear film motor [43] as a promising candidate for robotic application, due to its light weight, large output force, and large displacement capability. Among several variants, which include both induction [47,48] and synchronous [43] types, this work focused on a synchronous type that exhibits the largest output force capability. As a control method for the actuator, research was also conducted on keeping the output force constant using a closed loop [41]. However, to control the actuator precisely according to the phase difference, it is necessary to fix the film on a rigid frame, which does not take advantage of the lightweight and flexible features of this actuator. Therefore, in this study, we considered the adoption of open-loop driving. Some electrostatic actuators are driven in a closed loop, such as that of Ge et al. [26,46], but there are only a few. In other electrostatic actuators, open-loop driving is favored for its simplicity, while in closed-loop driving, it is not easy to control the actuators with accurate feedback of the phase difference. Therefore, the actuators are often driven in an open loop [19,23–25,27,40,44,45].

When considering adapting electrostatic actuators to robots, packaging is an issue. The linear film motor consists of flexible printed circuit (FPC) films, which have mechanical flexibility. As the motor needs to be operated while immersed in a dielectric liquid, the films need to be encapsulated in an enclosure. To fully use the mechanical flexibility of the films, the enclosure should also be flexible. One example of flexible enclosures is shown in Fig. 1. In the photo, a flexible bag made of thin plastic film containing the FPC films is filled with dielectric liquid. In previous studies on electrostatic film motors, a dielectric liquid called Fluorinert (3M) was mainly used because it has low viscosity and is less oily (less sticky). However, the liquid is volatile and can expand the enclosure bag like a balloon, if it is left in the bag for days. The expanded bag disturbs the motor motion, and should be avoided for practical use.

With that in mind, previous studies tried to evaluate silicone oils as an alternative liquid [49,50]. In particular, the focus was on highly viscous grades of silicon oil, which are less flammable. The studies evaluated the maximum load as the motor's output force, at which the motor steps out in a synchronous operation. Highly viscous oils create a considerably large drag resistance during movement [51], and

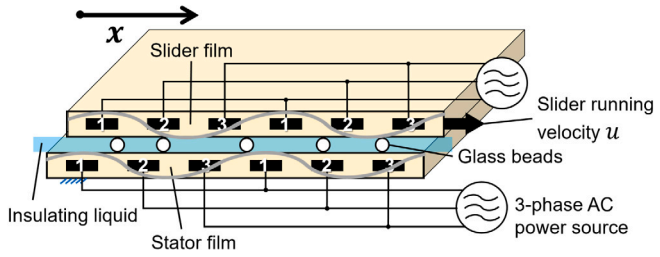



**Fig. 2.** Operation principle for synchronous electrostatic film motor using two sets of 3-phase AC voltages.

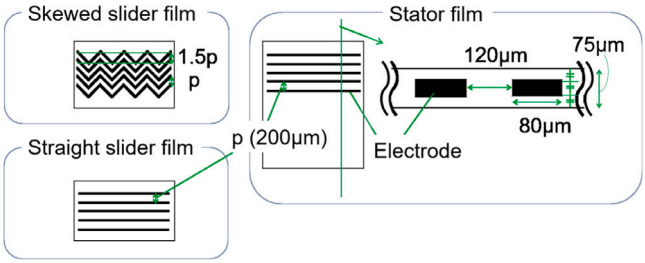



**Fig. 3.** Electrode pattern (skewed and straight) for the slider and the stator films (left) and its cross section (right).

therefore, one may expect that the output performance will decrease if a motor is driven in such a liquid. But surprisingly, studies have found that the output force while the motor was running does not change even though the films of the motor are moving within highly viscous silicone oil. However, the reason for this characteristic is still unclear.

This paper tries to clarify why the output force does not degrade in viscous oil even under dynamic situations in which the actuator speed exceeds 10 mm/s (as determined from experimental results in [50]), through simulations and experiments. The paper is structured as follows. The next section explains the principle of the synchronous electrostatic film motor and its force characteristics. Two different types, which have different electrode shapes, are described. Section 3 reviews the results of the previous studies regarding the effect of liquid viscosity on the output force. This section also examines a speculation raised in previous studies, and rejects it through some experimental observations. In Section 4, to explain the motor behavior in viscous liquid, including slider trajectories and output force, is simulated. The simulation results are verified experimentally in Section 5. The final section provides conclusions.

## 2. Synchronous electrostatic film motor

In this section, the structure of the electrostatic film actuator, the synchronous driving method, and the stepping out conditions are presented in detail. There are two types of electrostatic actuators, rotary and linear, and we will focus on the linear type.

### 2.1. Structures

The synchronous electrostatic film motor mainly consists of a stator film, a slider film, small glass beads, and an insulating liquid injected between the films, as shown in Fig. 2. The stator and the slider films have embedded three-phase electrodes, which are arranged with a regular pitch, $p$. In this paper, the pitch $p$ is 200 μm.

Whereas the stator electrodes have a straight shape, there are two different shapes for the slider electrodes, as shown in Fig. 3. One is a straight shape, the same as the stator, and the other shape is a zigzag. Use of the straight shape produces a larger peak force, but it also produces considerably large force ripple. The zigzag shape can average the force characteristics to reduce the force ripple by the skew effect [52], but the resulting peak force becomes smaller. In this paper, a motor using the straight slider electrodes is called the "straight type", and a motor using zigzag electrodes is called the "skewed type".

Regardless of type, the slider film is simply placed on the stator film in a typical motor configuration. To maintain a specific gap and avoid stiction due to the insulating liquid, 20-μm glass beads are sprinkled between the films. Although we did not strictly manage the amount of glass beads added, we observed under a microscope that there were about 10 beads per 3 pitch squares, so the volume ratio of glass beads to liquid was about 1%. We applied enough glass beads to make the actuator move smoothly, and fixed the film to the frame with screws to keep the ratio constant to some extent. Then, the gap was filled with dielectric liquid to allow application of a high voltage to the electrodes.

### 2.2. Operation

Among several different operation methods, this study used *dual-frequency operation* [43], in which two sets of 3-phase sinusoidal high voltages are applied to both the stator and the slider electrodes. Application of 3-phase voltages to 3-phase electrodes creates traveling voltage waves. The traveling speeds of the waves are proportional to the frequencies of the applied voltages. If the frequencies for the stator and the slider are $f_{st}$ and $f_{sl}$, respectively, the traveling speeds of their waves are $3pf_{st}$ and $3pf_{sl}$ relative to their electrodes. When a slider is moving at a speed of $u$, the absolute traveling speed of the slider wave is $3pf_{sl}+u$.

The slider is propelled by the electrostatic interaction between the waves. In a synchronous state, the two waves are engaged like a pair of gears; in other words, their absolute speeds become the same, $3pf_{sl}+u = 3pf_{st}$. This defines the synchronous speed as $u = 3p(f_{st}-f_{sl})$. Thus, the slider speed is proportional to the difference between the frequencies applied to the stator and the slider.

The actuator can be driven in a closed loop by sensing the phase difference precisely and adding a feedback voltage [41]. However, high-precision sensing is required, and parameters such as capacitance are easily changed in a bent FPC actuator. Therefore, in this study, the actuator was driven in a simple open loop. The driving speed can be controlled by the frequency of the voltage. In this study, a constant frequency was applied to create a constant synchronous speed to evaluate the effect of the liquid on the output force.

### 2.3. Thrust force characteristics and step-out

This study investigated the motor's output force from the behavior of its "step-out", that is, unwanted non-synchronous operation (as described in more detail at the end of this section). To provide preliminary knowledge, this subsection explains how step-out happens, using the motor's force characteristics.

Using phases $\phi_{st} = 2\pi f_{st}t$ and $\phi_{sl} = 2\pi f_{sl}t$, we define the 3-phase voltages for the stator and the slider, respectively, as

$$V_{st} = V\left[\sin(\phi_{st}), \sin(\phi_{st}-2\pi/3), \sin(\phi_{st}+2\pi/3)\right] \quad (1)$$

$$V_{sl} = V\left[\sin(\phi_{sl}), \sin(\phi_{sl}-2\pi/3), \sin(\phi_{sl}+2\pi/3)\right], \quad (2)$$

where $V$ is the voltage amplitude and $t$ represents time. Then, the motor's thrust force, $F_x$, is expressed as described in [52]:

$$F_x = \frac{3\pi}{2p}V^2[C_{m1}\sin(\phi+\theta_x) - 2C_{m2}\sin(\phi-2\theta_x) - 3(C_{l3}+C_{t3}+2C_{m3})\sin(3\theta_x)], \quad (3)$$

where $C_{m1}$ to $C_{m3}$ represent the amplitudes of the $i$th harmonics in the capacitance change between a slider electrode and a stator electrode, and $C_{l3}$ and $C_{t3}$ represent the amplitude of the capacitance variation between adjacent slider electrodes and adjacent stator electrodes, respectively. In a skewed type motor, the capacitance variations are negligible except for $C_{m1}$, which makes the second and third terms

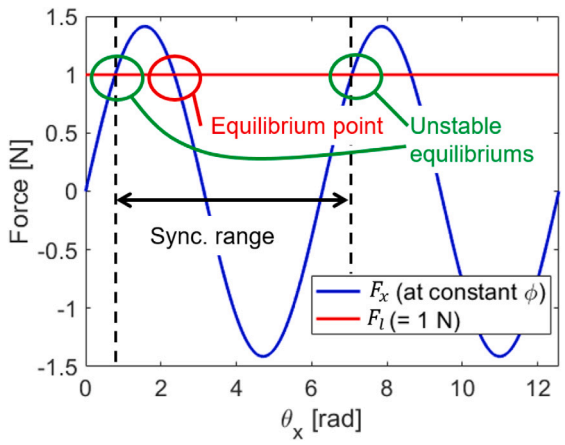



**Fig. 4.** Relationship between force and electric angle, $\theta_x$, in a skewed type motor. The graph shows the range in which the actuators can be driven synchronously. The blue curve shows the thrust force, $Fx$. The motor operates around the equilibrium point where the thrust force balances the external load, $Fl$.

in Eq. (3) disappear. The variable $\theta_x$ is an electric angle of the slider displacement $x$, and is defined as $\theta_x = 2\pi x/(3p)$. The angle $\phi$ denotes the phase difference between the two sets of 3-phase voltages, and is $\phi = \phi_{st} - \phi_{sl}$.

It should be noted that, in this paper, "thrust force" refers to the instantaneous electrostatic force expressed by Eq. (3). In contrast, the maximum external load given to the motor without causing step-out is defined as the "output force". The output force is measured at a constant synchronous speed and is thus a function of the speed.

Taking the characteristics of the skewed type as an example, Fig. 4 shows the thrust force, $F_x$, for a constant $\phi$. When $\phi$ is constant, the force changes sinusoidally against the slider position, $\theta_x$. If an external load, $F_l$, is applied to the slider, the motor will operate around the static equilibrium. In a dynamic situation, the operating point may move around the static equilibrium point. However, the operating point will be contained within the "sync range", which is one cycle of the sine wave between the two unstable equilibriums. If the phase difference $\phi$ changes with time, the equilibrium point and the corresponding sync range will also move. As a result, the slider will move to follow the motion of the sync range, which is synchronous operation.

If the external load exceeds the peak output force, there will be no equilibrium point and the operating point will drift away, which is called step-out, meaning the motor is out of synchronization. Even if the external load is smaller than the peak force, the operating point may drift away from the sync range for various reasons, such as inertia, viscosity, and distortions of the force characteristics. This also results in step-out. Especially in straight type motors, the latter step-out can happen more easily, as its force characteristics are considerably complicated and distorted.

## 3. Insulating liquid and its effect on actuation

In this section, the behavior of insulating liquids under a high voltage is presented based on previous studies and experimental observations.

### 3.1. Insulating liquid types

As shown in Eq. (3), the output force of the motor is proportional to the product of the applied voltage amplitude. Therefore, to obtain a large force, a high voltage is needed, typically more than 1 kV. Such a high voltage causes discharge to the surrounding air, especially within the gap between the stator and the slider. To prevent the discharge to the air, the motor is immersed in a dielectric liquid for high-performance operation [53].

**Table 1**
Comparison of insulating liquid properties.

| | Fluorinert | Silicone oil KF-96A | | |
|---|---|---|---|---|
| | FC-770 | 6cs | 50cs | 500cs |
| Viscosity [mPa s] | 1.44 | 5.55 | 48.0 | 485 |
| Viscosity [mPa s] (experiment) | 14.4 | 24.65 | 87.1 | 289 |
| Density [g/cm$^3$] | 1.8 | 0.925 | 0.960 | 0.970 |
| Relative permittivity | 1.9 (@1 kHz) | 2.60 (5cs) (@50 Hz) | 2.72 | 2.76 (@50 Hz) |
| Volatility [%] | 100 (cond. unknown) | <5 (3 h at 105 °C) | <0.5 | <0.5 (24 h at 150 °C) |
| Dielectric Strength [kV] | >40.0 (@2.5 mm) | >35.0 (5cs) (@2.5 mm) | >50.0 | >50.0 (@2.5 mm) |
| Flash point [°C] | None | 174 | >310 | >315 |

Many types of insulating liquids are used for electrostatic motors. For example, Niino et al. evaluated ester oil, silicone oil, and Fluorinert [54]. In more recent reports, Ge et al. evaluated Vertrel XF (Dupont), FR3 (Cooper Power Systems), and vegetable oils [26]. With regard to volatility, oils such as Fluorinert are a better choice than synthetic liquids. Oils can be classified into four categories [55], specifically mineral oil, silicone oil, synthetic esters, and vegetable oil. Among those categories, this work adopted silicone oil due to its low flammability.

As there are many different grades for silicone oil, this study selected grades by focusing on volatility, viscosity, and flash point. Generally, silicone oil with lower viscosity has a lower flash point, which makes it more likely to ignite in case of unexpected electric discharges. Therefore, from the safety viewpoint, higher viscosity is preferred. However, higher viscosity creates larger viscous friction and may degrade the motor performance, especially during dynamic operation, as described in Sindhu's paper [56]. In this study, we selected several different grades of silicone oil (Shin-Etsu). The grades chosen were 6cs, 50cs, and 500cs (these are grade names that represent their kinematic viscosities). They were chosen as they have relatively high flash points. Their key parameters are summarized in Table 1, together with those for Fluorinert FC-770, which has been used in some previous studies employing the same electrostatic motor.

### 3.2. Previous studies on the effect of liquid

Our previous studies focused on viscosity to evaluate the effects of insulating liquids on the output force. A straight type motor was used in those studies. Our first report [49] revealed that, although viscous oil created a large drag resistance, the measured output force for the motor was not degraded when the motor was driven in viscous oils.

In a recent study [50], more extensive experiments were carried out. This work also found that the output force did not decrease even when viscous oil was used. Specifically, even when using an oil with moderate viscosity, such as 48 mPa s, there was no significant drop in output force obtained by driving the actuator very slowly (1 mm/s) or by increasing the speed more than 10 times (10 mm/s, 20 mm/s, and 30 mm/s). This study also confirmed that, at least in the measured speed range, that the output force did not change much at different operation speeds, even though the drag resistance of the liquid is proportional to the speed.

### 3.3. Experimental observation of liquid flow

In the previous studies, even though the liquid created large viscous drag, the resulting output force was not lowered. To explain this, the previous studies speculated that the viscous liquid in the gap was actuated electrically, in a way similar to that for electrohydrodynamic pumps [57–59]. If the liquid itself is actuated within the gap, the effect of viscous resistance might vanish. However, in the previous studies, direct observation of the liquid flow within the gap was not attempted.

For a clear understanding of the motor behavior, this work confirmed whether the liquid is really flowing in the gap. For the observation, silicone oil 50cs was used. To track the oil motion, some of the oil was dyed using coloring powder (Orient Chemical, OIL YELLOW GG-S).

The results are shown in Fig. 5. In Fig. 5(a), the stator was arranged in the open air, and a silicone oil droplet was placed on the stator surface. Then, a high-voltage 3-phase current at 1.0 Hz was applied to the stator's electrodes. The picture clearly shows that the droplet moved in the traveling direction of the voltage wave. The location of the left edge of the droplet moved 1.8 mm in 3 s, which is equivalent to the traveling distance of the voltage wave. This result is consistent with a similar experiment in [50], and it shows that the silicone oil can be actuated by an electric field.

However, stators are not actually operated in open air or with oil surrounded by air. To observe the liquid flow under more practical conditions, the surface of the stator film was fully covered with silicone oil, into which a dyed oil droplet was injected. After that, the slider film was placed on top of the stator film to contain the oil, and voltages were applied to both the stator and the slider, such that the voltage waves traveled in the same direction in both the stator and the slider.

After 30 s of voltage application, the slider film was carefully removed and the location of the dyed oil was observed. The images in Fig. 5(b) show that, although the dyed oil spread during 30 s, its central position did not change. This means that under actual operating conditions, the oil within the gap does not move electrostatically. In other words, to actuate oil by an electric field, an air–oil interface is required. Because the air and oil have different permittivities, the interface receives an electrostatic force [60]. However, under actual motor operating conditions, the gap is completely filled with oil and the oil does not receive electrostatic force.

For a better understanding of this effect, another experiment was carried out as shown in Fig. 5(c). In this experiment, the stator surface was covered with silicone oil, in which glass beads were arranged. The electrodes in the stator film were subjected to 1-Hz high voltages for 1 s and the location of the glass beads was observed. The slider film was not used in this observation because placing and removing the slider film could change the location of the beads quite easily. The observation showed that the glass beads moved 600 μm, which is equivalent to the traveling distance of the voltage wave. This can be explained in the same way as for the experiment shown in Fig. 5(a). The dielectric constant of glass beads is not disclosed by the manufacturer, but based on the relative dielectric constant of glass, it is about 5.0. The relative dielectric constant of the oil used is about 2.7, so the glass bead constant is a little larger than that of the oil, which is thought to be the reason why the beads moved independently. The different permittivities of the glass and the surrounding oil created an electrostatic force, which moved the beads in a way similar to particle transport [61].

Glass beads are used in actual motor operation and they move electrostatically in the gap during operation. However, considering their small size (20 μm), the motion of the beads does not cause significant liquid flow in the gap. Therefore, we can say with high confidence that the silicone oil in the gap does not travel significantly due to the electrostatic effect.

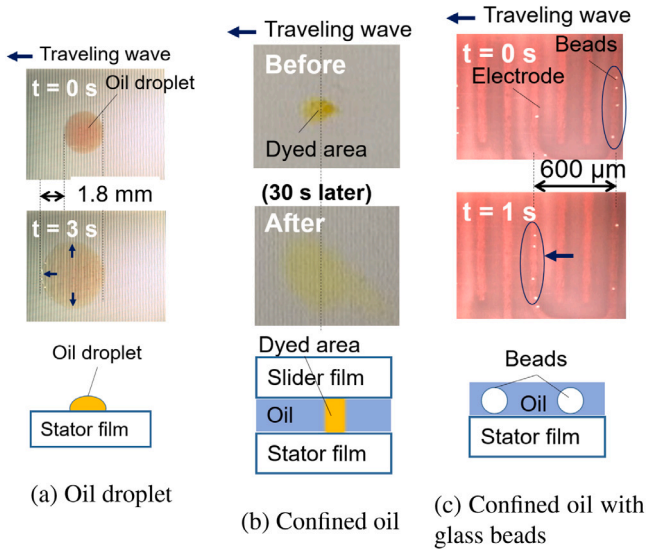



**Fig. 5.** Observation of the motions of oil and beads. (a) Oil droplet is directly placed on a film in open air. (b) Part of the oil confined between two films is dyed. (c) Glass beads are spread within the confined oil. The upper photos show the original state, whereas the lower photos show the state after voltage application. The voltage was 1.0 kV at 1.0 Hz and was applied along the $-x$ direction.

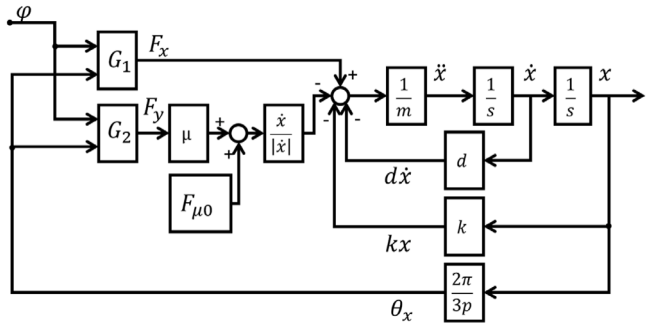



**Fig. 6.** Block diagram of the simulation model. The slider motion is simulated according to Eq. (6). The blocks $G_1$ and $G_2$ calculate $F_x$ and $F_y$ using Eqs. (3) and (4), respectively.

## 4. Simulations

As a possible cause of high output performance in highly viscous oil, the previous studies suggested the electrohydrodynamic effect. However, this mechanism was disproven in the observations described above. As an alternative cause, this study considered the interaction between the complicated force characteristics of the motor and the viscous resistance. To reveal such an interaction, slider motion was simulated using Matlab/Simulink (R2020a).

### 4.1. Simulation model

The simulation model is shown in Fig. 6. The model simulates the motion of the slider considering the Coulomb friction caused by electrostatic attraction, and the viscous drag caused by the liquid.

The thrust force was calculated using Eq. (3). To calculate the Coulomb friction, the electrostatic attraction force in the normal direction was calculated using the following force model proposed in [52]:

$$F_y = -\frac{9}{4y_0}V^2[2C_{m0} - C_{m1}\cos(\phi + \theta_x) + C_{m2}\cos(\phi - 2\theta_x) + 2C_{m3}\cos(3\theta_x)], \quad (4)$$

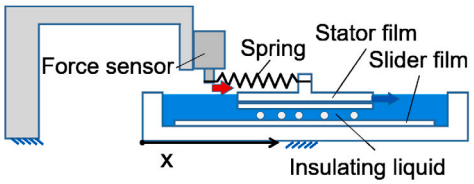



**Fig. 7.** Experimental setup for measuring output force using a spring. The slider pulls the spring connected to the force sensor until the motor steps out. Simulations employed this setup to identify the output force.

where $C_{m0}$ is the average capacitance between a slider electrode and a stator electrode. In our simulations, the gap $y_0$ was regarded as constant at 20 μm, which is the diameter of the glass beads. The adopted Coulomb friction model was the simplest one, where the friction is proportional to the normal attraction force. This represents the friction between the stator and slider films. As the experiment described below adopted a linear guide to support the slider, an additional constant friction was also applied ($F_{\mu 0} = 0.1$ N) to represent the friction of the guide. The viscous drag assumed a linear model where the drag is proportional to the speed of the slider. The damping coefficient, $d$, is given in unit of N s/m, and is related to the dynamic viscosity of the liquid, $\eta$, as

$$d = \frac{\eta S}{y_0}, \tag{5}$$

where $S$ is the overlapping area (55 cm$^2$, adopted from the experimental setup) between the slider and the stator.

In some simulations, it was assumed that a spring was connected to the slider to measure the maximum output force. In that case, a spring constant, $k$, was incorporated in the simulation model. The resulting equation of motion was

$$m\ddot{x} = F_x - (\mu F_y + F_{\mu 0})\frac{\dot{x}}{|\dot{x}|} - d\dot{x} - kx, \tag{6}$$

where $x$ is the slider displacement. In each simulation, the slider was given a constant synchronous speed. This speed was controlled by changing the phase difference, $\phi$, at a constant rate as

$$\phi = \frac{2\pi}{3p} v_s t, \tag{7}$$

where $v_s$ is the synchronous speed.

The capacitance parameters were measured from the experimental setup described below, with silicone oil 50cs injected within the gap. The values for the straight type are

$$\begin{cases} C_{m0} = 130, C_{m1} = 95, C_{m2} = 10, C_{m3} = 0.5 \\ C_{t3} = 1.0, C_{l3} = 1.0 \end{cases}$$

in units of pF, and those for the skewed type are

$$C_{m0} = 123, C_{m1} = 60.$$

The voltage amplitude, $V$, was fixed at 1 kV and the spring coefficient, $k$, was set to 0.016 N/mm when used; otherwise $k$ was zero.

### 4.2. Maximum output force

By activating the spring for force measurement, we identified the maximum output force under various conditions. The maximum spring force before the motor stepped out was identified as the maximum output force. This resembles the experiments described below, where the slider is connected to a spring and pulls it until the motor steps out, as shown in Fig. 7. In this simulation, the synchronous speed was changed between 1 and 30 mm/s.

Figs. 8 and 9 show the results for straight and skewed types, respectively. These plots evaluate the effect of liquid viscosity and

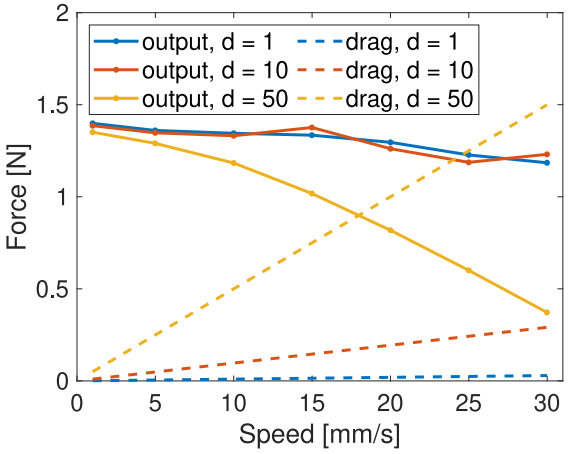



**Fig. 8.** Simulated output force for straight type (solid lines) and theoretical drag resistance (dashed lines) for different viscosities. The slider mass and the friction coefficient were $m = 0.05$ kg and $\mu = 0.04$, respectively.

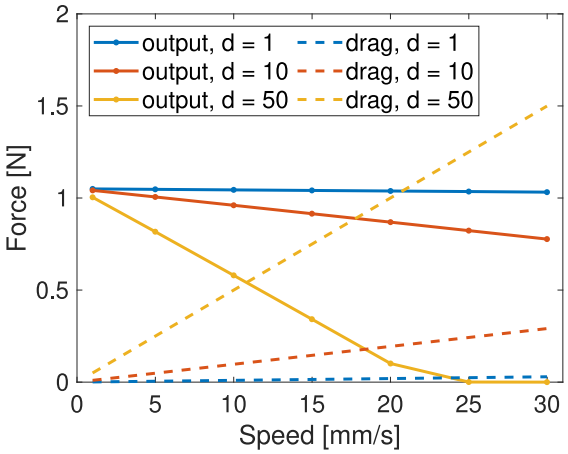



**Fig. 9.** Simulated output force for skewed type (solid lines) and theoretical drag resistance (dashed lines) for different viscosities. The slider mass and the friction coefficient were $m = 0.05$ kg and $\mu = 0.04$, respectively.

the synchronous speed. They show the identified output force at each speed, as well as the theoretical viscous drag calculated as $d\dot{x}$. Fig. 8 shows that, for the straight type, the output force variations for $d = 1$ and 10 are equivalent, even though the viscous drag for these viscosities is considerably different. The small output fluctuation found for $d = 10$ is likely due to numerical errors caused by the highly complicated force characteristics, as discussed below. For $d = 50$, the output force degraded rapidly as the speed increased. However, the loss of the output force was smaller than the theoretically calculated viscous drag. In contrast, for the skewed type in Fig. 9, the output force decreased linearly for all cases and the size of the decrease matched the estimated drag force.

To confirm that other parameters do not affect the results, the slider mass and the friction coefficient were also varied, as shown in Figs. 10 and 11. In these plots, the solid lines represent the results for the straight type motor, and the dashed lines represent those for the skewed type. Although a slight increase in output force was recognized for a small mass at a very low speed, the trend of the output force was not affected by those parameters.

The above results for the straight type confirmed the experimental findings in the previous studies, where the output force did not decrease as the theoretical viscous drag suggests. In contrast, there has been no experimental report on the effect of viscosity for the skewed type, and

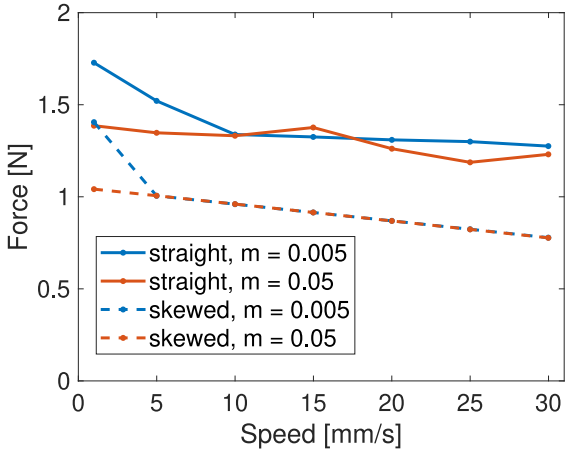



**Fig. 10.** Relationship between simulated output force and slider velocity when using straight type (solid line) and skewed type (dashed line) motors. The slider mass, $m$, was varied, while $d = 10$ and $\mu = 0.04$.

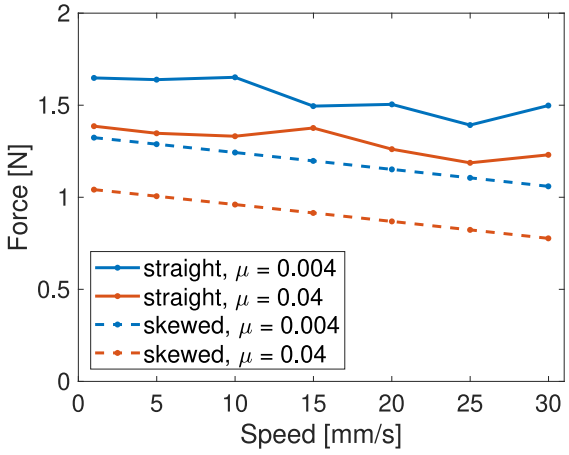



**Fig. 11.** Relationship between simulated output force and slider velocity when using straight type (solid line) and skewed type (dashed line) motors. The friction coefficient, $\mu$, was varied, while $m = 0.05$ and $d = 10$.

these simulations are the first trials to reveal its characteristics. The results show that, for the skewed type, the output force simply follows the theoretical calculation for the viscous drag.

### 4.3. Analysis of trajectories

To analyze why the effect of viscosity is not significant as the theoretical drag estimation, the motion of the slider was compared for different synchronous speeds and viscosities, using the straight type motor. Fig. 12 shows some examples of the slider motion, which are the motions at different synchronous speeds, 1 mm/s (which corresponds to the frequency difference, $f_{st} - f_{sl}$ of 1.66 Hz), 5 mm/s, and 20 mm/s. In these simulations, the mass of the slider, $m$, was set to 0.05 kg; the friction coefficient, $\mu$, was 0.04; and the damping coefficient, $d$, was set to 10, which corresponds to a dynamic viscosity of 36 mPa s. No load was given to the slider.

Fig. 13 plots the slider motion trajectories on the thrust force characteristics. The horizontal axis represents the phase difference, $\phi$, and the vertical axis represents the slider displacement, $\theta_x$. As $\phi$ was increasing at a constant rate during the simulation, the horizontal axis also represents the time elapsed in the simulation; however, the timescales are different among trajectories because the frequencies are different. The contour plot in the figure shows the thrust force

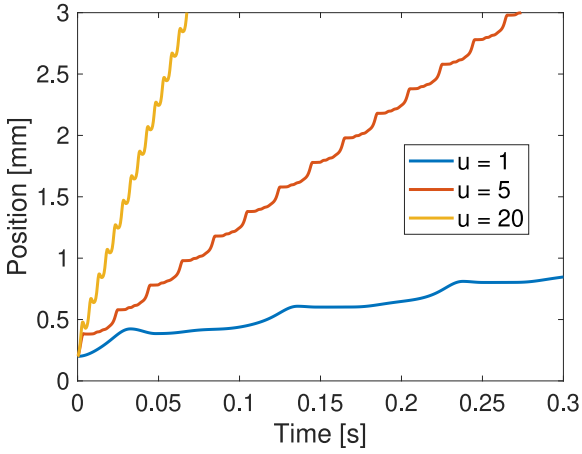



**Fig. 12.** Simulated slider trajectories for the straight type motor for different slider velocities ($m = 0.05$, $d = 10$, and $\mu = 0.04$).

characteristics calculated using Eq. (3). Then, the simulated motions at different synchronous speeds were overlaid on the force characteristics using bold lines.

As can be seen in the plots, the motion of the slider is repetition of "jump and stay". As the drag resistance is proportional to the instantaneous speed, the drag should mainly affect the "jump" period. If we focus on the "jump" motions in Fig. 13, we can find that the trajectories corresponding to the jump motions shift to the right as the synchronous speed increases. The shift means that the motion is delayed, which is probably caused by the viscous drag. Because of the shift, the trajectory approaches the force peak. As a result, the slider receives a larger thrust force that allows rapid acceleration to compensate for the drag resistance.

Also, due to the drag resistance, the proportion of the jump period within a motion cycle becomes larger. Therefore, we can also say that the speed change will simply affect the proportion of jump and stay periods, and does not cause step-out (which, in turn, defines the output force).

A similar tendency can be found in Fig. 14, which compares the trajectories for different viscosities, $d = 1$, $d = 10$, and $d = 50$. However, for the highest viscosity, the motion trajectory becomes almost linear due to the extremely large damping. In such a case, an almost constant drag force acts on the slider, and the output force simply decreases along with the increase of the speed, as can be seen in Fig. 8.

The above analyses suggest that the motion delay caused by viscous drag allows the full use of the peaks in the force characteristics, which compensates for the drag resistance. For the skewed type, however, the force characteristics are much smoother and the motor behavior is completely different. Fig. 15 shows the trajectories of the skewed type motor for different viscosities. The contour lines show the force characteristics. In contrast with the straight type motor, the force characteristics of the skewed type do not have distinct peaks and the resulting slider trajectories become straight lines. Therefore, the drag resistance is constantly applied to the slider. As the viscous drag increases either by higher viscosity or by higher speed, the trajectory shifts toward the right, but as there are no peaks in the force characteristics, the shift does not provide an advantage. Therefore, the output force for the skewed type simply decreases in accordance with the increase in the drag resistance.

## 5. Experiments

To verify the simulation results, the output force was experimentally measured for both straight and skewed motor types. Although the force for the straight type was measured in previous studies, the obtained

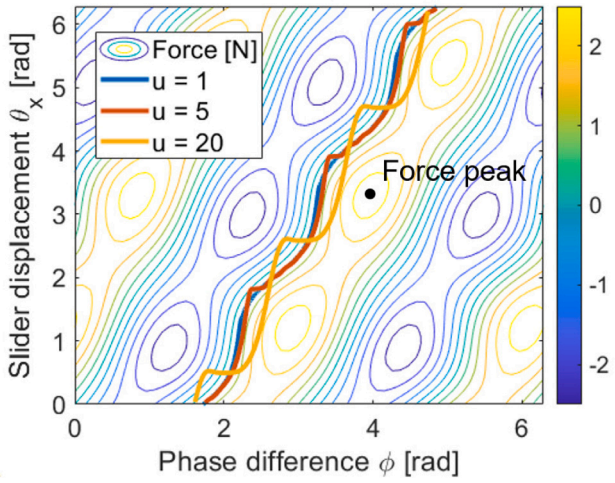



**Fig. 13.** Simulated slider trajectories plotted on thrust force characteristics (Eq. (3)) for the straight type motor. Contour lines represent the motor thrust force, and bold lines represent slider trajectories for three different speeds ($m = 0.05$, $d = 10$, and $\mu = 0.04$).

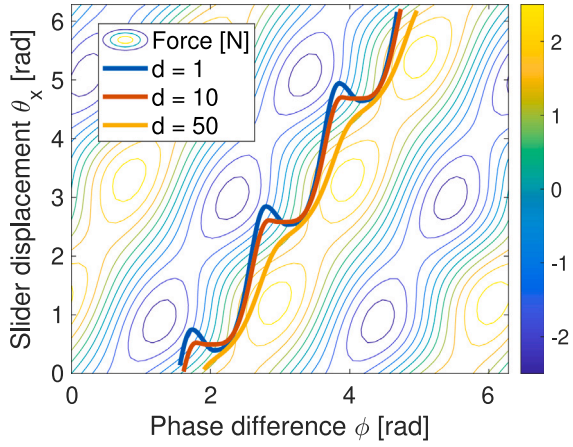



**Fig. 14.** Simulated slider trajectories for different viscosities plotted on force characteristics for the straight type motor ($u = 20$, $m = 0.05$, and $\mu = 0.04$).

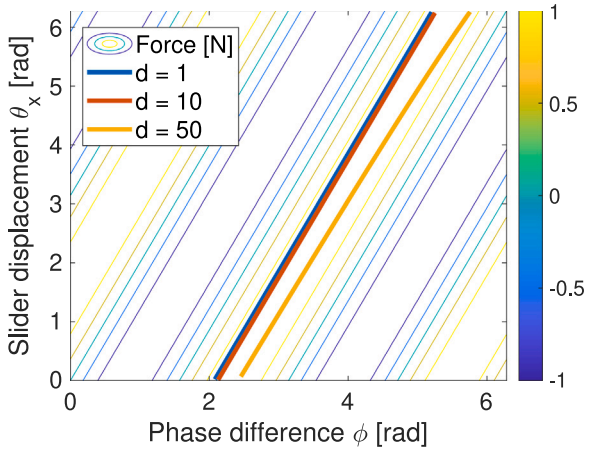



**Fig. 15.** Simulated slider trajectories for the skewed type motor for different viscosities. Contour lines show the force characteristics ($u = 20$, $m = 0.05$, and $\mu = 0.04$).

results were found to fluctuate, probably due to unstable operating conditions due to the slider arrangement. In the previous work, the slider film was attached to a plastic plate, which was simply placed

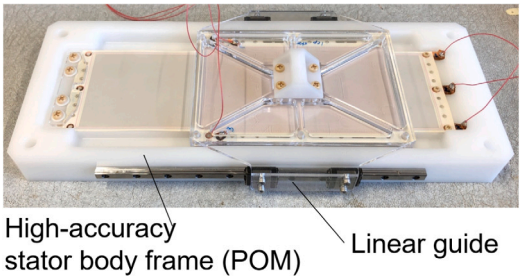



**Fig. 16.** Prototype motor used in the experiment. The slider motion is guided by a pair of linear guides.

on the stator. This might have caused the gap between the stator and slider to fluctuate. Therefore, in this work, we developed a different measurement setup, where the slider was supported by a linear guide, as shown in Fig. 16. As the slider film and the plastic plate were not rigid enough, compared to the electrostatic attraction force, the electrostatic attraction still directly affects the friction between the slider and stator films. However, the linear guide is helpful for maintaining a nearly constant gap. Also, it helps maintain the motion direction when an external load is applied, which is expected to improve the repeatability of the measurement.

### 5.1. Setup

The stator and the slider were formed from FPC films on which 3-phase electrodes were patterned. The electrode pitch, $p$, was 200 μm. The base and cover layers of the FPC films were liquid crystal polymer. The stator electrodes had a straight shape, whereas the slider had either straight or zigzag shapes. The size of the stator FPC film was 200 mm by 55.8 mm, while that of the slider was 110 mm by 75 mm. The effective actuation area, $S$, was 55 cm$^2$. The thickness of the FPC films was about 75 μm, as illustrated in Fig. 3.

In the experimental setup, the stator film was adhered to the body frame. The slider film was adhered to the plastic plate, which was supported by a pair of linear guides. Glass beads with a diameter of 20 μm were sprinkled in the gap between the stator and the slider. Then, the gap was filled with one of four dielectric liquids: Fluorinert FC-770 (3M) and three different grades of silicone oil (Shin-Etsu Chemical Co., Ltd.; KF-96A-6cs, 50cs, and 500cs).

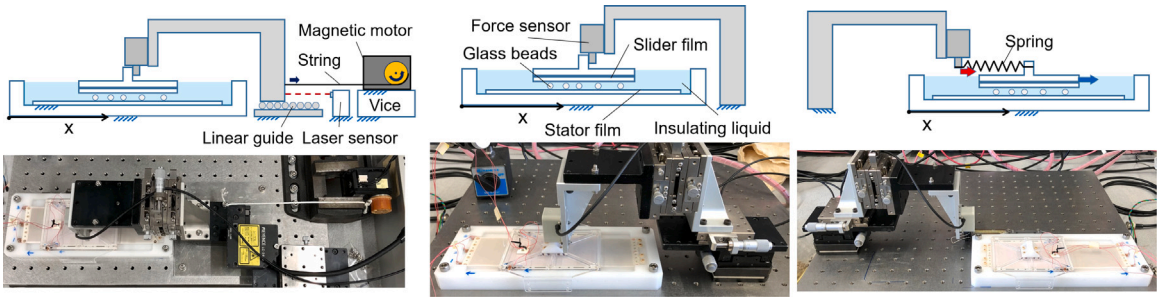
Two sets of 3-phase sine-wave voltages were generated by two multi-channel oscillators (HIOKI, 7075) and were amplified 1000 times by 6 high-voltage amplifiers (NF Corp., HV-4321). The body frame of the setup was fixed to the table, and the slider plate was connected to a load cell (Kyowa, LVS-2KA) directly or through a spring.

### 5.2. Measurement of viscous drag

Before measuring the output force for the electrostatic motor, we identified the viscous drag caused by the dielectric liquid in the actual motor setup. No voltage was applied to the motor, and the viscous drag acting on the slider was measured by simply pulling the slider with an external motor.

To facilitate that, the slider plate was directly connected to the load cell, and an electromagnetic motor (Robotis, Dynamixel MX-64) pulled the load cell at an almost constant speeds of roughly 1 mm/s, 10 mm/s, 20 mm/s, and 30 mm/s [see Fig. 17(a)]. The slider position was monitored using a laser displacement sensor (Keyence, LC-2440) and the slider was pulled for about 80 mm.

The results are shown in Fig. 18. In addition to the four liquids, the force without any liquid was also measured as the baseline. The base friction was found to be 0.55 N. The theoretical lines were plotted, including this base friction. The results show an almost linear tendency




(a) Drag resistance measurement (b) Static force measurement (c) Dynamic force measurement

**Fig. 17.** Cross section (upper) and appearance (lower) of experimental setup for force measurements. In this experiment, the apparatus was set up to measure the thrust force under three different conditions.

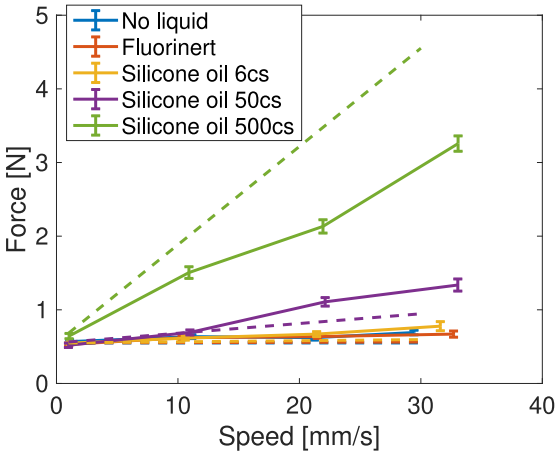



**Fig. 18.** Viscous drag measured using the prototype motor. No voltage was applied to the motor, and the slider was moved by an external motor. The error bars represent standard deviations during movement of 80 mm. The solid lines represent the measurement results, and the dashed lines represent the theoretical values.

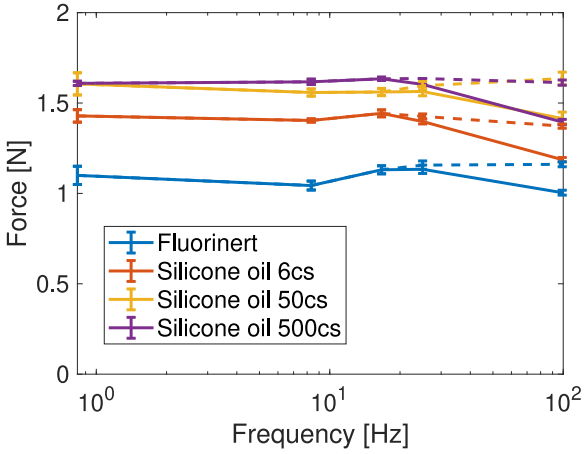



**Fig. 19.** Measured static output forces for different insulating liquids. The same 3-phase voltages were applied to the stator and slider (in the same direction), resulting in zero synchronous speed. The voltage amplitude was $V = 1.0$ kV. Error bars represent standard deviations of 10 experiments. Dashed lines show compensated values, considering the loss of voltage amplification of the amplifiers.

against the pulling speeds, but the slopes of the plots showed a slightly deviation. This deviation could be attributed to the fact that the gap between the slider and stator was not constant, but values were on the same order as the theoretical lines estimated from the nominal viscosity of the liquids.

### 5.3. Measurement of static output force

Our observations in Section 3 concluded that there is no significant liquid flow caused by electrostatic force. To further confirm this, the output force at zero speed was measured under different speeds of the voltage traveling waves. In this experiment, the same 3-phase voltages were applied to both films of a skewed type motor, such that the waves traveled along the same direction in both the stator and slider. The resulting speed difference, as well as the slider speed, was 0 m/s.

The slider was directly connected to the load cell and some displacement was introduced to change the phase difference, $\phi$. This created an almost sinusoidal force variation, and its peak value was recorded as the output force [see Fig. 17(b)]. The applied frequencies were 0.83 Hz, 8.33 Hz, 16.67 Hz, 24.99 Hz, and 100 Hz. If the liquid in the gap was actuated by the traveling voltage waves, the different frequency would cause different liquid speeds, and in turn, different output forces.

The results are shown in Fig. 19. In these experiments, the signal generator generated sine waves with an amplitude of 1 V, which corresponded to a nominal applied voltage of 1.0 kV. However, it was confirmed that the amplification ratio was slightly lower in the high-frequency range. Therefore, the voltage drop was compensated theoretically, as shown by dashed lines in the figure.

The dashed lines show that the force did not change for different applied frequencies. This confirmed again that the liquid inside the gap is not actuated by the electrostatic force, or at least, it does not affect the output force.

Because the motor was re-assembled every time the liquid was changed, the measurement conditions, especially the gap length, might have been slightly different among the measurements with different liquids. This probably caused the different output forces for different silicone oils. However, in general, the forces for silicone oils were found to be larger than that for Fluorinert, probably due to the higher permittivity of the silicone oils.

### 5.4. Output force

Finally, the output force of the motor was measured in a dynamic condition, where the slider was actuated at a constant speed against a spring connected to the load cell, as shown in Fig. 17(c). As the motor was operated, the slider pulled the spring until it stepped out from the synchronization state. The maximum force before the step-out was recorded. The synchronous speed was changed among the measurements, by varying the applied frequencies as ±0.83 Hz, ±8.33 Hz, ±16.67 Hz, and ±24.99 Hz, which correspond to 1 m/s, 10 m/s, 20 m/s, and 30 m/s, respectively.

Three liquids were evaluated, which were Fluorinert FC-770 and silicone oils 6cs and 50cs, whose damping coefficients, $d$, are roughly

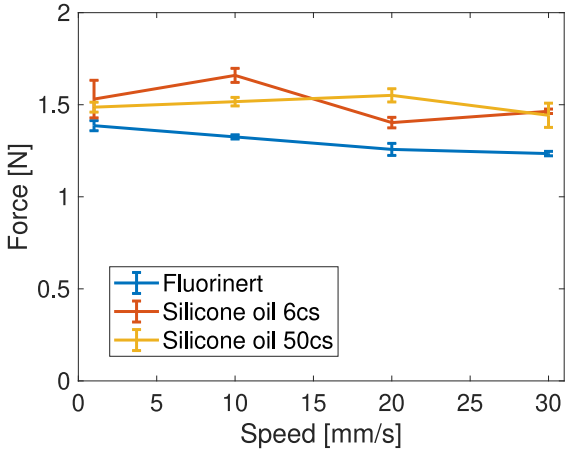



**Fig. 20.** Output force for straight type motor measured at various synchronous speeds. The voltage amplitude was fixed at $V = 1.0$ kV. Error bars represent standard deviations for 10 experiments.

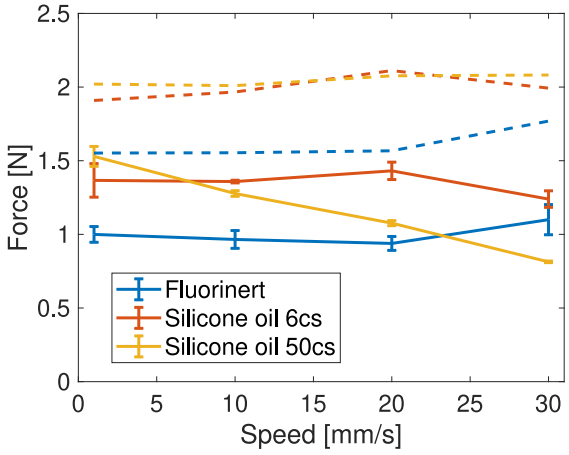



**Fig. 21.** Output force measured for the skewed type motor (solid lines). Dashed lines show the sum of the measured output force and the viscous resistance values in Fig. 18. The dashed lines are almost constant regardless of speed.

0.4, 1.5, and 13 N s/m, respectively. Silicone oil 500cs (equivalent to $d \approx 133$) did not produce stable motion and was thus omitted from the measurement. The measurement was done for both straight and skewed type motors. Figs. 20 and 21 show the results for the straight and skewed types, respectively. For the straight type, although some variations were observed, the output force was almost constant for the three liquids regardless of the operating speed. In contrast, for the skewed type, whereas the output force remained almost constant for low-viscosity liquids (Fluorinert and silicone oil 6cs), the output degraded almost linearly for the high-viscosity liquid (silicone oil 50cs). Dashed lines in this plot represent the sum of the output force and drag resistance, which were found to be almost constant regardless of operation speed. This clearly indicates that the reduction of the output force in the skewed type is simply due to the drag resistance of the liquid, as the simulation suggested.

### 5.5. Discussion

The experimental results generally supported the simulation results in Section 4. The important aspect of the obtained results is the difference between the types of motor. For the straight type, the output force remained almost the same for different viscosities and different speeds in the measured ranges. In contrast, for the skewed type, the output force decreased in a very simple manner. The decrease of the output force was found to be almost equivalent to the drag force caused by the insulating liquid.

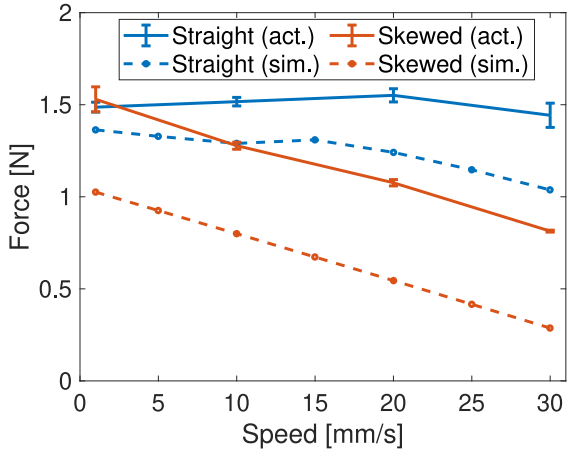



**Fig. 22.** Comparison between the experimental results (solid lines) using silicone oil 50cs, and the simulation results for the corresponding viscous coefficient (dashed line). The experimental data were taken from Figs. 20 and 21.

For quantitative comparison with the simulation, the output force was simulated using the viscosity of silicone oil 50cs, for both the straight and skewed motor types. The results are shown in Fig. 22. The experimental results for silicone oil were taken from Figs. 20 and 21 for comparison. Although there are offsets between the simulation and experiment, their trends are in good accordance for both motor types. The offset probably came from the measurement error in the gap length and/or the motor capacitance. Therefore, we can conclude that the unique output force behavior of the electrostatic motor is a result of the interaction between the complicated force characteristics and the viscous friction. Further, it only appears in the straight motor type.

## 6. Conclusions

In this study, a synchronous electrostatic film motor was used to investigate the effect of the insulating liquid on the output force through simulations and experiments. Previous studies found that the output force of the motor did not decrease even when the motor was operated in a viscous dielectric liquid. However, the previous studies did not make the reason clear. The results of this study revealed three important findings.

The first is that the insulating liquid in the gap of the motor is not actuated electrostatically during motor operation. The previous studies speculated that the liquid might be actuated electrostatically and affect the output performance. Our observation, however, clearly disproved this speculation.

The second finding is that the reason for the non-degraded output force with viscous liquid is the existence of force peaks in the force characteristics of the motor. Because these force characteristics are complicated, the motor shows step-wise motions, which consist of "jump and stay" actions. When the slider is subjected to a heavy drag resistance, the jump action is delayed to obtain the peak force that can compensate the drag resistance. Also, the drag resistance slowed the speed of the jump action within the step-wise motion, but it simply changes the proportion between jump and stay in a step motion cycle. As a result, the output force is not significantly affected by the drag resistance.

Finally, it was found that this unique behavior appears only in the straight type motor. This is because the force peaks described above are suppressed in the skewed type motor. Therefore, a skewed type motor does not show the unique behavior with a viscous liquid.

The use of dielectric liquid is necessary for all high-performance electrostatic motors, not just for the motor discussed in this paper. Generally, a dielectric liquid with lower viscosity creates less drag resistance, but it could cause a packaging problem due to higher volatility, or a safety problem due to a lower flash point. The findings of this work pave the way to using high-viscosity liquids for high-performance electrostatic motors. Although the force peaks may not exist in other electrostatic motors, the results of this work will provide an insight for those working on different electrostatic motors.

## CRediT authorship contribution statement

**Masahiko Osada:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Visualization, Supervision, Project administration. **Guangwei Zhang:** Methodology, Software, Validation. **Fernando Carneiro:** Writing – review & editing. **Masaaki Muromachi:** Methodology, Supervision. **Shunsuke Yoshimoto:** Writing – review & editing, Supervision. **Akio Yamamoto:** Formal analysis, Investigation, Resources, Writing – review & editing, Supervision, Project administration.

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